

L-Seg: An end-to-end unified framework for multi-lesion segmentation of fundus images



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ABSTRACT

Diabetic retinopathy and diabetic macular edema are the two leading causes for blindness in working-age people, and the quantitative and qualitative diagnosis of these two diseases usually depends on the presence and areas of lesions in fundus images. The main related lesions include soft exudates, hard exudates, microaneurysms, and haemorrhages. However, segmentation of these four kinds of lesions is difficult due to their uncertainty in size, contrast, and high interclass similarity. Therefore, we aim to design a multi-lesion segmentation model. We have designed the first small object segmentation network (L-Seg) that can segment the four kinds of lesions simultaneously. Taking into account that small lesion regions could not response at high level of network, we propose a multi-scale feature fusion method to handle this problem. In addition, when considering the cases of both class-imbalance and loss-imbalance problems, we propose a multi-channel bin loss. We have evaluated L-Seg on three fundus datasets including two publicly available datasets - IDRiD and e-optha and one private dataset - DDR. Extensive experiments have demonstrated that L-Seg achieves better performance in small lesion segmentation than other deep learning models and traditional methods. Specially, the mAUC score of L-Seg is over 16.8%, 1.51% and 3.11% higher than that of DeepLab v3+ on IDRiD, e-optha and DDR datasets, respectively. Moreover, our framework shows competitive performance compared with top-3 teams in IDRiD challenge. The source code of L-Seg is available at: <https://github.com/guomugong/L-Seg>.

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1. Introduction

Diabetic retinopathy (DR) and its complication diabetic macular edema (DME) have been the leading causes for blindness in working-age individuals. DR has become a serious public health problem in the world. The World Diabetes Foundation estimates that there will be 438 million people with DR by the year 2030 [1]. However, studies have shown that diabetic screening and effective interventions at early stages can significantly reduce the incidence of DR-induced blindness [2]. Therefore, it is crucial to diagnose DR at an early clinical stage. Since lesions in retinal fundus are an important indicator of DR, analyzing retinal fundus images is the most popular method for DR screening. In clinical practice, ophthalmologists usually diagnose and grade DR by the presence of microaneurysms (MA), soft exudate (SE), hard exudate (EX), and haemorrhage (HE) (see Fig. 1). Specially, the presence of hard

exudate is the main indicator to diagnose DME. However, the examination of fundus images is time-consuming and small lesions are hard to observe even for experts with rich experience. Therefore, developing an automatic method for lesion segmentation is of great importance.

Although many efforts have been made in recent studies for lesion detection, it is still a challenging task due to the following aspects. (1) The structure of lesions is complex, and there are huge differences between lesions in different local areas in terms of shape, size and intensity. The first challenge is to build a model that can describe complex lesions. (2) The size of most of lesions is very small. For example, some haemorrhages, hard exudates and microaneurysms are only one to several pixels in size, making it easy to be confused with noises. The second challenge is to segment small objects from retinal fundus images with complex structures, such as vessels and other noises. (3) There is high inter-class similarity for different lesions in retinal fundus. For example, it is also difficult for human eyes to distinguish soft exudates and hard exudates, or to distinguish haemorrhages and microaneurysms. The

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Fig. 1. An example of fundus image from the IDRiD dataset (a) retinal fundus image; (b) fundus image with annotation (red markings denote EX, green markings denote HE, blue markings denote MA, and yellow markings denote SE).

third challenge is to reduce the confusion rate for similar lesions. (4) It is not uncommon that some non-lesion structures in fundus images look similar with lesions, such as reflective vessels and drusen. The fourth challenge is to distinguish lesions from other similar structures or noises. (5) Great difference usually exists in retinal fundus images obtained by different imaging devices, which increases the difficulty in detection of some lesions such as microaneurysms [3]. The fifth challenge is to build a flexible and robust model for different devices.

Benefiting from big data and the improvement of computing capacity, many deep learning-based works focus on DR screening. However, most of their works are just running a classification network and crucial diagnostic evidence cannot be provided [2,4,5], which is harmful to communication between doctors and patients. Therefore, it is necessary to locate lesions. Recently, deep learning-based methods have been applied to lesion detection, such as hard exudate segmentation [6,7]. To the best of our knowledge, there have been no reports on using a single framework to segment HE, SE, MA, and EX simultaneously. Most of the existing lesion segmentation models only consider one kind of lesion which makes it hard to meet clinical requirements since a single kind of lesions cannot determine DR levels. In this paper, we propose an end-to-end unified framework called L-Seg, for simultaneous segmentation of multiple lesions. When a fundus image is fed into L-Seg, four segmentation probability maps are simultaneously generated, each of which corresponds to one kind of lesion. Furthermore, we propose a multi-channel bin loss (MCB) for segmentation of small lesions to avoid both class-imbalance and loss-imbalance problems. We have evaluated L-Seg on two publicly available datasets - IDRiD [8] and e-ophtha [9] and one private dataset - DDR. Experimental results show that L-Seg with MCB loss outperformed state-of-the-art segmentation models including HED, FCRN, CASENet and DeepLab v3+, and traditional EX detection methods. For the IDRiD dataset with four kinds of lesions, L-Seg with MCB loss achieved the best performance in all kinds of lesions, and its mAUC is 5.77%, 16.92%, and 16.86% higher than that of HED, CASENet and DeepLab v3+ on IDRiD dataset, respectively. For the DDR dataset, L-Seg with MCB loss achieved the highest AUC scores in EX, SE and MA, and its mAUC value is 3.11% and 13.01% higher than that of DeepLab v3+ and HED. For the e-ophtha dataset with only two kinds of lesions, L-Seg with MCB loss achieved the best performance compared with baseline models in MA segmentation and mAUC. Further, we compared L-Seg with the systems of the top 10 IDRiD challenge teams, and L-Seg with MCB loss also showed competitive performance over these state-of-the-art systems.

Our main contributions are as follows:

- We proposed an end-to-end unified framework called L-Seg, for simultaneous segmentation of multiple lesions. A weighted fusion module is introduced to enhance the learning ability of

the model, so that L-Seg can process information of a complex, multi-lesion structure.

- We proposed a MCB loss for segmentation of small lesions to solve both class-imbalance and loss-imbalance problems, so as to avoid the trend of either misclassifying small lesions as backgrounds or misclassifying backgrounds as lesions.
- We conducted extensive experiments on two publicly available datasets and one private dataset. Ablation studies have demonstrated that both weighted fusion and MCB loss are beneficial to improving the performance of multi-lesion segmentation. And comparative studies have revealed that L-Seg with MCB loss outperformed the state-of-the-art segmentation models and showed competitive performance compared with the state-of-the-art systems in IDRiD challenge.

The remainder of this paper is organized as follows. Related works are introduced in Section 2. The details of L-Seg are described in Section 3. Experiments and analysis are given in Section 4. Discussion and Conclusions are drawn in Sections 5 and 6.

2. Related Works

Fundus lesion segmentation is essentially a problem of semantic segmentation. Therefore, we have analyzed the related works about fundus lesion segmentation, as well as semantic segmentation.

2.1. Fundus lesion segmentation

A number of methods have been proposed in recent literatures for lesion segmentation of fundus images. Here we divide lesion segmentation methods into three categories from the aspect of Machine learning, including Un-Machine learning based methods (manually designed features + manually designed classification rules) [3,10–12], Semi-Machine learning based methods (manually designed features + learnable classification model) [13–15], and Full-Machine learning based methods (learnable features + learnable classification model, e.g., deep learning) [6,7,16,17]. Imani and Pourreza [11] proposed a pipeline to segment hard exudates. They remove vessel trees from fundus image firstly and then a thresholding algorithm is adopted to get exudate candidates. This kind of thresholding-based method only considers the bright feature of exudates so that it is hard to generalize to real world application. While, deep learning based methods can learn much richer feature representations and are suitable to clinical application. From a more general point of view, it is clear that many hyperparameters are needed for Un-Machine learning methods, and as a result, such methods behave poorly when used in another dataset and increase the workload of personnel. Compared with Un-Machine learning based methods, Semi-Machine learning based methods can benefit from a larger training set by training a model for classification.

Table 1

Statistical information of IDRiD, e-ophtha and DDR datasets, where # means the percentage of lesions in all pixels.

Dataset	#EX	#HE	#MA	#SE	Total images
IDRiD	0.90%	1.01%	0.10%	0.19%	81
e-ophtha	0.20%	–	0.01%	–	21
DDR	0.19%	0.33%	0.02%	0.05%	757

However, the features are still designed by human in these methods. It is difficult to design features that are most suitable for the applications.

Full-Machine learning based methods can automatically extract the most suitable features and train the model over the training set. It is anticipated that the performance is increased as the training set becomes larger. As one of the Full-Machine learning based methods, deep learning has been popular in recent years due to its outstanding performance, such as computer vision [18–20] and medical image analysis [2,6,17]. Tan et al. [17] designed a 10-layer pixel-wise deep convolutional neural network to automatically segment hard exudates, hemorrhages, and microaneurysms. Each pixel is classified by the statistical cues of a patch centered at the pixel, making such method time-consuming and difficult to meet real-time clinical requirements since fundus images are always large in size. Mo et al. [6] designed an image-level fully convolutional residual network (FCRN) for hard exudate segmentation. The FCRN can produce an exudate probability map for a fundus image with only one forward pass, which is efficient. However, the FCRN is a single-lesion segmentation model, and as a result, it can only provide limited help to DR and DME diagnosis.

In this work, we designed an end-to-end deep model for multi-lesion segmentation. Considering the characteristics of fundus lesions, two problems should be noticed in model design. One problem is lesion sizes that vary considerably, and the other one is the complex and un-regular structure of lesions. The popular multi-scale cues, which have been used for many models, are employed in the present study to handle lesions of different sizes. Furthermore, we propose a weighted fusion module to enhance the learning ability of the model, so that information contained in complex lesions can be mined and processed by the model.

2.2. Semantic segmentation

As an important task in computer vision, deep learning-based semantic segmentation has been well-studied in recent years, and demonstrated remarkable performance. Most of the existing semantic networks [19,21] focused on large-sized or normal-sized objects such as persons, dogs and cats. However, fundus lesions are usually small-sized, and some of them are even one to several pixels in size, which may cause the best semantic segmentation model [19] to exhibit poor performance in segmentation of small lesions. In consideration of the characteristics of small-sized lesions, two problems should be noticed in model design. One problem is information loss caused by successive pooling operation, and the other one is class-imbalance problem due to a low ratio of lesions to backgrounds as shown in Table 1.

Dilated convolution [22] and capsule networks [23] were proposed for avoiding information loss by discarding the pooling operation, and multi-scale [21] was proposed for capturing small-sized objects by low-level features. In this work, we used a multi-scale model structure since multi-scale can capture lesions of all sizes with the features at different levels.

A common method for addressing class-imbalance problem is to introduce a weight factor β for hard exudate pixels and $1 - \beta$ for background, i.e., class-balanced loss [18]. However, class-balanced loss leads to another problem, i.e., background pixels tend to be misclassified as lesions since the weighted loss for

misclassifying a background pixel is much smaller than that for misclassifying a lesion pixel, which is called a loss-imbalance problem here as it is introduced by the weighted loss. Obviously, such loss-imbalance problem should be more serious for small-sized objects due to the lower ratio of object pixels to background pixels. As shown in Table 1, lesion pixels are much smaller than background ones, so that background pixels with smaller weighted loss tend to be misclassified as lesions with larger weighted loss. Therefore, it tends to generate a noisy result that contains many false lesions by using the class-balanced loss, which greatly reduces the performance of fundus lesion segmentation. In practice, a large part of background pixels is easy to classify due to the large difference between these background pixels and the lesion pixels. In this work, we proposed a MCB loss that focuses on hard-to-classify background pixels, and only the top-k background pixels with high loss are considered in each iteration of model training. Ablation studies have demonstrated that the MCB loss shows a large performance improvement in fundus lesions segmentation, capable of effectively handling both class-imbalance and loss-imbalance problems.

3. L-Seg

L-Seg was designed as an end-to-end framework to simultaneously segment multiple kinds of lesions from a fundus image by only one pass inference, which has three characteristics including: deep supervision for avoiding gradient vanish problem, multi-scale for different sizes of lesions and weighted fusion for multi-lesion segmentation. Furthermore, two loss functions, i.e., multi-channel loss and MCB loss, were proposed for L-Seg, and both of them are suitable for multi-class cases. Compared to multi-channel loss that only considers class-imbalance, MCB loss can be used to handle both class-imbalance and loss-imbalance problems.

3.1. Architecture

The architecture of the L-Seg is shown in Fig. 2, which consists of the following parts:

- Feature learning: VGG16 [24], pre-trained on ImageNet, was used as the backbone network for L-Seg. VGG16 consists of 5 groups of convolution layers and three fully connected layers. With the deepening of network, the receptive field of each convolution layer increases. Thus, the learned features are from low-level with more details to high-level with more structures. For L-Seg on multi-lesion segmentation, we cut off the pool5 layer and all the fully connected layers of VGG16, and the conv1, conv2, conv3, conv4 and conv5 in Fig. 2(a) correspond to 5 groups of convolution layers in VGG16, respectively.
- Side feature extraction and deep supervision: The last one in each group of convolution layers is connected to a convolution layer with a kernel size 1×1 and a channel size c , where c denotes the number of lesion types. To make the size of an output feature map the same as that of an input image, a deconvolution layer is used for up-sampling feature maps. For each side feature map, a loss layer is connected for avoiding gradient vanish problem by deep supervision. The structure of side feature extraction modules is shown in Fig. 2(c). It should be noted the up-sampling is not needed for side 1 feature map since side 1 feature map is of the same size as the input image.
- Weighted fusion: To enhance the learning ability of the L-Seg, the feature maps at different scales are combined by a weighted fusion module instead of the simple Eltwise-Sum. As shown in Fig. 2(b), the fusion weights are different for each kind of lesion, and they are automatically learned through the end-to-end training.

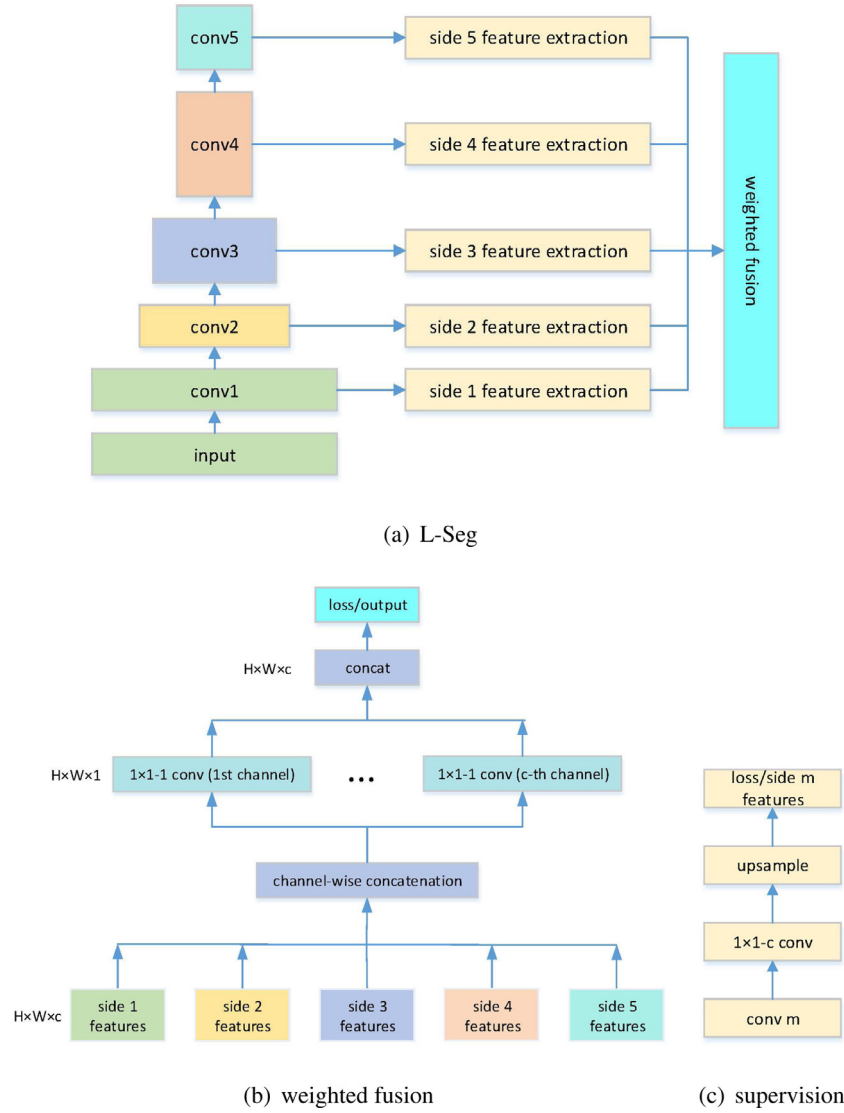


Fig. 2. (a) An overview of L-Seg architecture. The input of the model is an RGB color fundus image, and the outputs are segmentation probability maps. Each kind of lesion has one segmentation probability map. (b) Weighted fusion module. The input in this method is five side features (size: $H \times W \times c$) obtained at different side layers. The output is a fused segmentation result, $H \times W \times c$ in size. (c) Side feature extraction and deep supervision. c denotes a type of lesion. The value of c is 4 and 2 on IDRid dataset and e-optha dataset, respectively.

- **Loss layers:** There are six losses to be optimized in the L-Seg, including five deep supervision losses and one output loss. Each loss corresponds to a stand-alone segmentation model. The input to the loss layer is a feature map and a corresponding target segmentation result, both of them having the size of $H \times W \times c$, in which H and W denote the height and width of input image respectively and c represents the number of lesion types. The parameters of L-Seg are learned by back-propagating these losses.

All parameters of L-Seg are automatically learned from labeled samples by the end-to-end training. We will describe side feature extraction, deep supervision, weighted fusion and loss function in detail here.

3.2. Side feature extraction and deep supervision

We add supervision information after the last convolutional layer of each stage in VGG16 based on the following two aspects. On one hand, supervision information is helpful for alleviating the

gradient vanish problem [25]. On the other hand, once deep supervision is added, there exists five stand-alone segmentation models each with different scale in L-Seg. For small lesion point with only a few pixels, such as MA, it is hard to response at stage 4 or stage 5, therefore, watching the segmentation result at stage 3 or stage 2 is meaningful. The output of each side feature extraction operation can be regarded as the middle segmentation result, and once they are fed into the weighted fusion operation, the final segmentation result is obtained.

As shown in Fig. 2(c), the procedure of deep supervision and side feature extraction can be divided into three parts. First, the last convolutional layer of each stage in VGG16 is connected to a convolution layer with kernel size 1×1 and channel number c . This procedure can be regarded as feature map generation. Then, up-sampling operation is performed to restore the size of the feature map to the input image so that we can calculate the loss value. In our implementation of L-Seg, we use bilinear interpolation to up-scale. One thing worth noting is that for the first side feature extraction, up-sampling is not necessary, since the size of feature map is the same with input image, which means no pool-

ing operation or convolution operation with big stride was performed at former level. Third, the feature map is connected to a loss function as we will describe below. The six losses are optimized simultaneously, not one at a time, in the training phase.

3.3. Weighted fusion

As shown in Fig. 2(b), the weighted fusion module in L-Seg consists of channel-wise concatenation, feature fusion and concat operations.

Channel-wise concatenation: Five feature maps are fed into the weighted fusion module, denoted as s_1, s_2, s_3, s_4 and s_5 , from left to right, respectively, each having the size of $H \times W \times c$. Each one in c channels of s_i represents a feature map for one kind of lesion. And for each kind of lesion, we fuse five feature maps by channel-wise concatenation. This procedure can be formulated as below.

$$\text{ConcatChannel}_k = \text{concat}(s_1^k, s_2^k, s_3^k, s_4^k, s_5^k) (1 \leq k \leq c) \quad (1)$$

where k denotes the k th channel of s_i . After channel-wise concatenation, c feature maps (ConcatChannel_k) are generated, each having the size of $H \times W \times 5$.

feature fusion: ConcatChannel_k are connected to a 1×1 convolution layer. The value of convolutional kernel corresponds to fusion weights. This procedure can be formulated as below.

$$\text{fuseChannel}_k = \sum_{i=1}^5 w_k^i \text{ConcatChannel}_k^i (1 \leq k \leq c) \quad (2)$$

where w_k^i denotes fusion weight, and ConcatChannel_k^i means the i th channel of ConcatChannel_k .

concat: The last step of weighted fusion module is to concat fuseChannel_k having the size of $H \times W \times 1$. As a result, a model output having the size of $H \times W \times c$, named $\text{side}^{(\text{fuse})}$, is generated.

3.4. Loss function

As shown in Table 1, the distribution of lesion pixels and background pixels is highly unbalanced, leading to the popular class-imbalance problem that tends to misclassify lesion pixels as background ones. Weight-balanced sigmoid cross entropy loss [18] can handle the class-imbalance problem, but it is only suitable for binary classification. Multi-label loss function [26] is proposed for multi-class cases, but it is not suitable for multi-lesion segmentation since there should be only one label for each pixel in a fundus image. Here we propose a multi-channel loss function, meeting the characteristics of multi-lesion segmentation. Furthermore, as an improved version of multi-channel loss function, MCB loss is proposed for avoiding both class-imbalance and loss-imbalance problems.

3.4.1. Multi-channel loss function

Given an image X with the label Y , each pixel value of Y denotes its class, in which 0 means background and 1 to c means different kinds of lesions, respectively. The multi-channel sigmoid cross entropy loss (MC Loss) can be formulated as:

$$\begin{aligned} L(X, Y; W) &= \sum_{k=1}^c l_k(X, Y; W) \\ &= \sum_{k=1}^c \sum_{i=1}^{H \times W} \{-\beta I(k, i) \log p(y_k^{(i)} = 1|X; W) \\ &\quad - (1 - \beta)(1 - I(k, i)) \log(p(y_k^{(i)} = 0|X; W))\} \end{aligned} \quad (3)$$

where W denotes model parameters, and H and W denote the height and width of both image X and the corresponding label Y ,

respectively. β is the percentage of background pixels in the image, taken as the weight factor to handle class-imbalance problem. $p(y_k^{(i)} = 1|X; W)$ represents the probability of a pixel $i \in X$ being a lesion pixel with the class label k , outputted by the model. $I(k, i)$ is an indicator function that describes whether the label of the pixel i is k , as defined in Eq. (4).

$$I(k, i) = \begin{cases} 1, & Y^{(i)} = k \\ 0, & \text{otherwise} \end{cases} \quad (1 \leq k \leq c) \quad (4)$$

3.4.2. Multi-channelbin loss function

The MC loss deals with class-imbalance problem by weight factor β . However, the background pixels tend to be misclassified as lesion pixels in MC loss since the loss for a misclassified background pixel is much smaller than that for a misclassified lesion pixel, i.e., loss-imbalance problem. In the MC loss, all foreground and background pixels are considered. In practice, a large part of background pixels is easy to classify due to the large difference between these background pixels and lesion pixels. Therefore, we propose multi-channel bin loss (MCB Loss), an improved version of MC loss, to handle the loss-imbalance problem by focusing more on hard-to-classify background pixels, and only the top- k background pixels with high loss are considered in each iteration of model training. The formula of MCB loss, similar to that of MC loss (Eq. (3)), is defined below:

$$\begin{aligned} L_B(X, Y; W) &= \sum_{k=1}^c l_k(X, Y; W) \\ &= \sum_{k=1}^c \sum_{i=1}^{H \times W} \{-\beta I(k, i) \log p(y_k^{(i)} = 1|X; W) \\ &\quad - (1 - \beta) I_B(k, i) \log(p(y_k^{(i)} = 0|X; W))\} \end{aligned} \quad (5)$$

The calculation of $I(k, i)$ is the same as Eq. (3). In the MCB loss, $I_B(k, i)$ divides the background pixels into two parts, pixels that counts to loss and others. $I_B(k, i)$ is defined as below:

$$I_B(k, i) = \begin{cases} 1, & i \in B_k \\ 0, & \text{otherwise} \end{cases} \quad (6)$$

where B_k is the set of background pixels with the top λY_+ losses (i.e., the hard-to-classify pixels) for k -th lesion, where Y_+ is the number of lesion pixels and λ is a hyper-parameter. As a result, the weight factor β can be calculated as follows:

$$\beta = \frac{\frac{1}{c} \sum_{k=1}^c B_k}{Y_+ + \frac{1}{c} \sum_{k=1}^c B_k} \quad (7)$$

Now, the sum of five side deep supervision losses is defined as below:

$$L^{(\text{side})}(X, Y; W) = \sum_{m=1}^5 L_B^{(m)}(\text{side}^{(m)}, Y; W) \quad (8)$$

where $\text{side}^{(m)}$ is the segmentation result of the m th side feature extraction. We have obtained the fusion result $\text{side}^{(\text{fuse})}$ by weighted-fusion module, so that the output loss is defined as below:

$$L^{(\text{fuse})}(X, Y; W) = L_B(\text{side}^{(\text{fuse})}, Y; W) \quad (9)$$

At last, putting everything together, we aim to minimize the following loss function by back-propagation algorithm:

$$W^* = \arg \min (L^{(\text{side})}(X, Y; W) + L^{(\text{fuse})}(X, Y; W)) \quad (10)$$

4. Experiments and analysis

4.1. Datasets

We evaluated our method over two publicly available datasets, i.e., IDRiD and e-optha, and one private dataset DDR.

Table 2
Implementation details of baseline models.

Model	Framework	Official code?	Pre-trained	Dataset	Learning rate	Learning rate policy	Max iteration	Batch size
HED	caffe	Yes	ImageNet	IDRiD	1e-8	step	25000	1
			ImageNet	DDR	1e-8	step	25000	1
			ImageNet	e-ophtha	1e-8/1e-7	step	20000	1
FCRN	caffe	No	ImageNet	IDRiD	1e-6	step	2000	10
			ImageNet	DDR	1e-6	step	4000	10
			ImageNet	e-ophtha	1e-6	step	1600	10
CASENet	caffe	Yes	ImageNet	IDRiD	1e-8	step	50000	1
			ImageNet	DDR	1e-8	step	90000	1
			ImageNet	e-ophtha	1e-8	step	30000	1
DeepLab v3+	TensorFlow	Yes	Pascal VOC	IDRiD	1e-4	poly	30000	1
			Pascal VOC	DDR	1e-4	poly	100000	1
			Pascal VOC	e-ophtha	1e-4	poly	30000	1

The IDRiD (Indian Diabetic Retinopathy Image Dataset) [8] is available for the segmentation and grading of retinal image challenge hosted by ISBI (International Symposium on Biomedical Imaging) conference 2018. The dataset consists of 81 images, each of which has a resolution of 4288×2848 , and pixel-level annotations of EX, HE, MA and SE are provided. It is the only one dataset that provides four types of lesions annotations. The partition of training set and testing set are provided on IDRiD, with 54 images for training and the rest 27 images for testing. All images in the testing set have EX, MA and HE, and 14 images have SE.

The e-ophtha dataset [9] is a publicly available dataset that consists of two parts: e-ophtha EX and e-ophtha MA. There are 82 retinal images in e-ophtha EX, of which 47 images contain hard exudates and 35 images are normal images. The e-ophtha MA dataset consists of 148 images with MA and 233 normal images. We selected 21 images with both hard exudates and microaneurysms from the e-ophtha dataset. In our experiments, we used 11 images for training and the rest 10 images for testing.

DDR is a general high-quality dataset collected and labeled by us for diabetic retinopathy screening, lesion segmentation and lesion detection. It contains 13,673 fundus images from 147 hospitals, covering 23 provinces in China. These images were captured using 42 types of fundus camera with a 45-degree field of view, and all images have been desensitized for common use. For segmentation task, 757 fundus images are provided with pixel-level annotation for EX, HE, MA and SE. Among them, 601 images had HE, 570 images had MA, 239 images had SE and 486 images had EX. This dataset contains 383 images for training, 149 images for validation and the rest 225 images for testing. In our experiments, we trained models using training set and validation set, i.e., 532 images, and evaluated our method on testing set.

The statistical information of these three datasets is summarized in Table 1, which suggests that the distribution of lesion pixels and background pixels is highly unbalanced for all three datasets. Specially, the percentage of MA on e-ophtha dataset is ten times smaller than that on IDRiD dataset. And, the percentage of EX, HE, MA and SE on DDR dataset is at least three times smaller than that on IDRiD dataset.

4.2. Baseline segmentation models

We compare L-Seg with single-lesion segmentation model including FCRN [6] and HED [18], and multi-lesion segmentation model including CASENet [26] and DeepLab v3+ [19]. FCRN was designed for hard exudate segmentation specifically, and it achieved superior performance compared with many traditional methods [11,27,28] on hard exudate segmentation. However, it is only suitable for single-lesion segmentation. In [6], FCRN was trained using the frequency-based balance loss [29], whereas we use weighted sigmoid cross entropy loss here. HED was first pro-

posed for edge detection, but our experiments show it achieves competitive performance compared with FCRN, so that we used it as a baseline model. HED contains five side-output layers with different receptive fields, and each can be regarded as a stand-alone segmentation model. The weight-fusion layer combines the segmentation result of five side-output layers together to get a final segmentation result. Since HED is only suitable for single object segmentation, thus we trained multiple HED models, each model used for single-lesion segmentation. DeepLab v3+ used to achieve the state-of-the-art performance in semantic segmentation task of the PASCAL VOC 2012. It was designed for segmenting normalized objects, such as persons, dogs and cats. In the present study, we use DeepLab v3+ for multi-lesion segmentation to show that the state-of-the-art semantic segmentation model for normal objects is difficult to achieve good performance in small object segmentation. CASENet is an end-to-end deep semantic edge segmentation architecture. Different from HED that could only detect edge and background pixels, CASENet is category-aware and actually a multi-class edge segmentation model.

4.3. Evaluation metrics

Same to the IDRiD challenge [8], we used AUC, i.e., the area under the Precision-Recall (PR) curve, as evaluation metric. In our experiments on IDRiD dataset, the PR curve was obtained by thresholding the results for 0 to 1 with an interval of 0.1, keeping in line with the IDRiD challenge. So that it is possible to compare L-Seg with IDRiD teams. When it comes to other two datasets, we used 256 thresholds to get more accurate AUC values. In addition, we employed mean AUC (mAUC), the mean value of AUC of different kinds of lesions, to evaluate overall performance of the model in multi-lesion segmentation. Just like the definition of mAP on object detection task [30].

4.4. Implementation details

We will describe the data augmentation methods and the training details of L-Seg and baseline models in this section. Table 2 shows an overview of parameter settings of baseline models.

4.4.1. Model Implementation

We trained and tested L-Seg using the *caffe* library [31]. We implemented a CPU and GPU version for multi-channel loss function and a CPU version for multi-channel bin loss function. We used official implementation of HED, CASENet and DeepLab v3+. HED and CASENet were implemented on *caffe*, and DeepLab v3+ was implemented on TensorFlow. FCRN was implemented by us, since their code is not available online.

Table 3

Statistical results of ablation experiments over e-ophtha datasets (best results are shown in bold).

Model	e-ophtha		
	EX	MA	mAUC
L-Seg (Eltwise-Sum, MC)	0.0980	0.0470	0.0725
L-Seg (Eltwise-Sum, MCB)	0.1523	0.1639	0.1581
L-Seg (Weighted-Fusion, MC)	0.2382	0.1101	0.1742
L-Seg (Weighted-Fusion, MCB)	0.4171	0.1687	0.2929

4.4.2. Initialization

In our experiments, we initialize the convolution blocks of VGG in HED and L-Seg with models pretrained on ImageNet dataset [32]. FCNR was fine-tuned from ResNet-50 [20] pretrained also on ImageNet. And we initialize the weight of CASNet with models ResNet-101 [20] pretrained on ImageNet. DeepLab v3+ is pretrained on Pascal VOC dataset.

4.4.3. Hyper-parameters

For training L-Seg, we performed SGD with a batch size of 1, a momentum of 0.9, and a weight decay of 0.0005 on all three datasets. We set the learning rate, step size, gamma and max iteration to $1e-9$, 15000, 0.1, and 25000 for IDRiD. On e-ophtha dataset, learning rate was initialized to $1e-8$ and the max iteration was set to 20000. On DDR dataset, the step size and max iteration was set to 30000 and 45000, respectively, and the setting of other hyper-parameters is the same with those of on IDRiD dataset. Moreover, λ in MCB loss is set to 10 on IDRiD and DDR datasets, and it is set to 20 on e-ophtha dataset.

4.4.4. Data preprocessing and augmentation

The size of training images is too large to fit them into GPU memory, therefore, we resize each image and its ground-truth to 1440×960 (width $\times$ height) before training on IDRiD and e-ophtha datasets. The images from DDR dataset are scaled to much lower resolution with width 1000.

Data augmentation method was applied to generate more training samples. We used various transformations to augment the training set, including rotation by an angle of 90, 180 and 270, flipping horizontally and vertically. The training images of three datasets were augmented by a factor of 5. So that, there are 324, 66 and 3192 training images on IDRiD, e-ophtha and DDR datasets, respectively. We use the same amount of data to train HED, FCNR, CASNet and L-Seg. Besides our data augmentation method, the official implementation of DeepLab v3+ has extra augmentation method, so that it uses more training samples.

4.5. Results

4.5.1. Ablation study

Weighted-Fusion and MCB Loss. To show the effectiveness of weighted fusion and MCB loss, we performed comparative experiments with different combinations by fusion methods (i.e.,

Table 5

AUC scores of side features and fusion features in L-Seg on IDRiD dataset.

Feature	EX	HE	MA	SE
side 1	0.4569	0.0345	0.0028	0.0184
side 2	0.6802	0.0531	0.2332	0.1634
side 3	0.7263	0.3230	0.3983	0.4324
side 4	0.6712	0.5572	0.2838	0.5897
side 5	0.6195	0.5996	0.1294	0.6787
fusion	0.7945	0.6374	0.4627	0.7113

weighted fusion and commonly used Eltwise-Sum) and loss functions (i.e., MC loss and MCB loss). Eltwise-Sum operation [20,21] is a popular method to fuse multi-scale or multi-level features, where the input features are summed by element to get the fused features. Each input feature map has the same contribution to the output. While our proposed weighted fusion can assign different weights to inputs, so that it is much flexible and accurate compared with Eltwise-Sum. The results are shown in Tables 3 and 4.

We can observe from Table 4 that compared with Eltwise-Sum, weighted fusion led to significant improvement of AUC on the e-ophtha dataset with two kinds of lesions. Specially, the mAUC is improved by at least 10%, regardless of MC loss or MCB loss being used. And, compared with MC loss, MCB loss gets much higher AUC in MA and EX. Those observations show that weighted fusion and MCB loss is suitable for segmentation two types of lesions.

When the number of lesion types was increased to 4 on the IDRiD and DDR datasets, the same improvement of mAUC can also be found, as can be seen in Table 3. Specially, weighted fusion led to 2.03% improvement of mAUC on IDRiD when MCB loss was employed. There is still 1.59% improvement of mAUC when MC loss was used on IDRiD dataset. In addition, MCB loss achieved a much higher mAUC value compared with MC loss on IDRiD, no matter weighted fusion or Eltwise-Sum was used. Specially, MCB loss led to 2.17% improvement on IDRiD. The improvement is much significant on DDR dataset, weighted fusion led to at least 20% improvement of mAUC for both MC and MCB loss. And, MCB loss achieves 4.39% improvement when weighted fusion was used. The metrics on the DDR dataset is lower than that in IDRiD, since the distribution of lesion points is much more unbalanced on the DDR.

Compared to L-Seg with Eltwise-Sum and MC loss, L-Seg with weighted fusion and MCB loss led to 3.76% improvement on IDRiD, 22.04% improvement on e-ophtha, and 25.52% improvement on DDR, verifying the effectiveness of weighted fusion and MCB loss.

Side Features. To validate the side features, as well as the deep supervision losses, we summarize the results produced by each side feature at different scales in Table 5. We can observe some interesting results. First, combining segmentation from five scales yield better performance and this means all five side features make their own contribution to the performance gain. Second, for MA segmentation, side 3 features achieve the highest performance among five side features. This indicates deeper network is unnecessary for MA segmentation, since it is only a few pixels wide.

Table 4

Statistical results of ablation experiments over IDRiD and DDR datasets, ES denotes Eltwise-Sum, WF denotes Weight-Fusion, MC denotes Multi-channel loss and MCB denotes Multi-channel bin loss (best results are shown in bold).

Model	IDRiD					DDR				
	EX	HE	MA	SE	mAUC	EX	HE	MA	SE	mAUC
L-Seg(ES, MC)	0.7410	0.5808	0.4710	0.6629	0.6139	0.1076	0.1167	0.0162	0.0218	0.0656
L-Seg(ES, MCB)	0.7671	0.5998	0.4692	0.6889	0.6312	0.1760	0.2159	0.0375	0.0530	0.1206
L-Seg(WF, MC)	0.7707	0.6084	0.4313	0.7086	0.6298	0.4903	0.3255	0.0937	0.1980	0.2769
L-Seg(WF, MCB)	0.7945	0.6374	0.4627	0.7113	0.6515	0.5546	0.3586	0.1052	0.2648	0.3208

Table 6

Statistical results of baseline models and L-Seg over IDRiD and DDR datasets (best results are shown in bold).

Model	IDRiD					DDR				
	EX	HE	MA	SE	mAUC	EX	HE	MA	SE	mAUC
HED [18]	0.7723	0.5084	0.4271	0.6673	0.5938	0.4241	0.1919	0.0496	0.0972	0.1907
FCRN [6]	0.5472	0.4200	0.3383	0.5154	0.4552	0.1996	0.1255	0.0134	0.0454	0.0960
CASENet ([26])	0.7564	0.4462	0.3992	0.3275	0.4823	0.2777	0.2625	0.1005	0.1304	0.1928
DeepLab v3+ ([19])	0.7118	0.4772	0.1514	0.5912	0.4829	0.5349	0.3894	0.0223	0.2120	0.2897
L-Seg	0.7945	0.6374	0.4627	0.7113	0.6515	0.5546	0.3586	0.1052	0.2648	0.3208

Table 7

Statistical results of baseline models and L-Seg over e-optha datasets (best results are shown in bold).

Model	e-optha		
	EX	MA	mAUC
HED [18]	0.1345	0.0639	0.0992
FCRN [6]	0.1675	0.0269	0.0972
CASENet ([26])	0.1715	0.1565	0.1640
DeepLab v3+ ([19])	0.5512	0.0045	0.2778
L-Seg	0.4171	0.1687	0.2929

4.5.2. Comparison with baseline models

The AUC scores for each dataset are summarized in Tables 6 and 7. And the PR curves of these models on three datasets are shown in Figs. 3–5. In addition, we selected the point of the PR curve with the highest f1score as threshold to binary the probability segmentation map. We visualize the binary segmentation maps for fundus images from IDRiD and DDR test sets in Figs. 6 and 7. We did not visualize the binary segmentation maps on e-optha test set, since it contains only two kinds of lesions.

We can observe from Fig. 6 that HED recognized SE, EX and HE well, but there exists serious mis-classification on MA segmentation, since it cannot handler loss-imbalance problem. DeepLab v3+ could recognize large SE areas, but it could only recognize a few numbers of MA, since small MA will not response at higher levels. DeepLab v3+ shows poor performance in very small lesion segmentation, i.e. MA. There are many mistakes between EX and SE segmentation in CASENet which means it behaves poorly in similar lesion segmentation. Moreover, there is a large part of false positive hard exudates in FCRN. L-Seg could not only recognize small MAs but also large SE, EX areas. We can observe from the binary segmentation maps on DDR test set (see Fig. 7) that there still exists serious mis-classification of HED on MA recognition. FCRN failed to segment some obvious HE and EX areas, and the MA areas are very coarse. Moreover, the HE areas of FCRN, and DeepLab v3+ are much coarser compared with L-Seg. DeepLab v3+ still behaves poorly in MA segmentation.

We can observe from Tables 6 and 7 that L-Seg achieves the highest AUC scores on EX, MA and SE segmentation compared with baseline models on both IDRiD and DDR datasets, and it gets the highest mAUC score on all three datasets. When L-Seg is compared to the single-lesion segmentation models, it resulted in a mAUC 5.77%, 13.01% and 19.37% higher than that of HED on IDRiD, DDR and e-optha datasets, respectively. And the gap between L-Seg and FCRN is very large. When L-Seg is compared to the multi-lesion segmentation models, it resulted in a mAUC score 16.92% and 16.86% higher than that of CASENet and DeepLab v3+ on IDRiD dataset, respectively. And the mAUC score of L-Seg was 12.89% and 1.51% higher than that of CASENet and DeepLab v3+ on e-optha dataset, respectively. L-Seg still shows superior performance compared with CASENet and DeepLab v3+ on DDR dataset, the mAUC score of L-Seg is 3.11% higher than that of DeepLab v3+.

The experimental results demonstrate the superior performance of L-Seg over HED, FCRN, DeepLab v3+ and CASENet in multi-lesion segmentation.

Table 8

Comparison with top ten teams in IDRiD challenge.

Model (Team)	EX	HE	MA	SE	mAUC
VRT (1st)	0.7127	0.6804	0.4951	0.6995	0.6469
PATech (2nd)	0.8850	0.6490	0.474	–	–
iFLYTEK-MIG (3rd)	0.8741	0.5588	0.5017	0.6588	0.6484
SOONER (4th)	0.7390	0.5395	0.4003	0.5369	0.5539
SAIHST (5th)	0.8582	–	–	–	–
lzyuncc_fusion (6th)	0.8202	–	–	0.6259	–
SDNU (7th)	0.5018	0.4572	0.4111	0.5374	0.4769
CIL (8th)	0.7554	0.4886	0.3920	0.5024	0.5346
MedLabs (9th)	0.7863	0.3705	0.3397	0.2637	0.4401
AIMIA (10th)	0.7662	0.3283	0.3792	0.2733	0.4367
L-Seg	0.7945	0.6374	0.4627	0.7113	0.6515

4.5.3. Comparison with IDRiD challenge teams

The comparison between L-Seg and the top 10 teams on IDRiD competition is shown in Table 8. Compared with other models, the competitive performance of L-Seg lies in three aspects. First, when considering segmentation of four kinds of lesions together, L-Seg achieved the highest mAUC value. Second, L-Seg ranked No. 1 on SE segmentation, No. 3 on HE segmentation, No. 4 on MA segmentation, and No. 5 on EX segmentation. Third, for the top 3 teams that gave conference reports to describe their models, they used different network architectures for different lesion segmentation tasks, in which a large number of hyper-parameters need to be adjusted in training stage, and four models are needed to detect lesions in testing stage. For other teams, they did not describe their models, and their results have a big gap with those of the top 3 teams. For top 3 teams, although they did not address all types of lesions, they did well on single-lesion segmentation. And, our proposed L-Seg is the only end-to-end unified framework that generates multi-lesion segmentation results and shows competitive performance compared with the top 3 teams.

4.5.4. Comparison with traditional methods

Besides comparing with deep learning methods, we also compare L-Seg with two traditional methods on e-optha EX dataset. E-optha EX dataset contains 47 images with exudates and 35 healthy images. In our experiments, we only use 47 unhealthy images just as that did in [27]. Also, only 21 images out of 47 images have MA. So that we train L-Seg for EX and MA segmentation. In our experiments, we adopt the 5-fold cross validation to evaluate the performance of L-Seg. For fair comparison, we use sensitivity, positive prediction value (PPV) and F-score to evaluate the EX segmentation result just as that did in [11] and [27]. The calculation method of metrics is the same as that in [27], we both use component-level validation rather than pixel-level validation. We can observe from Table 9 that our L-Seg outperforms other two traditional methods. Specially, L-Seg achieves the highest Sensitivity, PPV and F-score among all methods which means our methods can detect more EX areas with higher precision. Furthermore, traditional methods require complicated preprocessing and post-processing steps, while our L-Seg works in an end-to-end manner, thus the process is simplified.

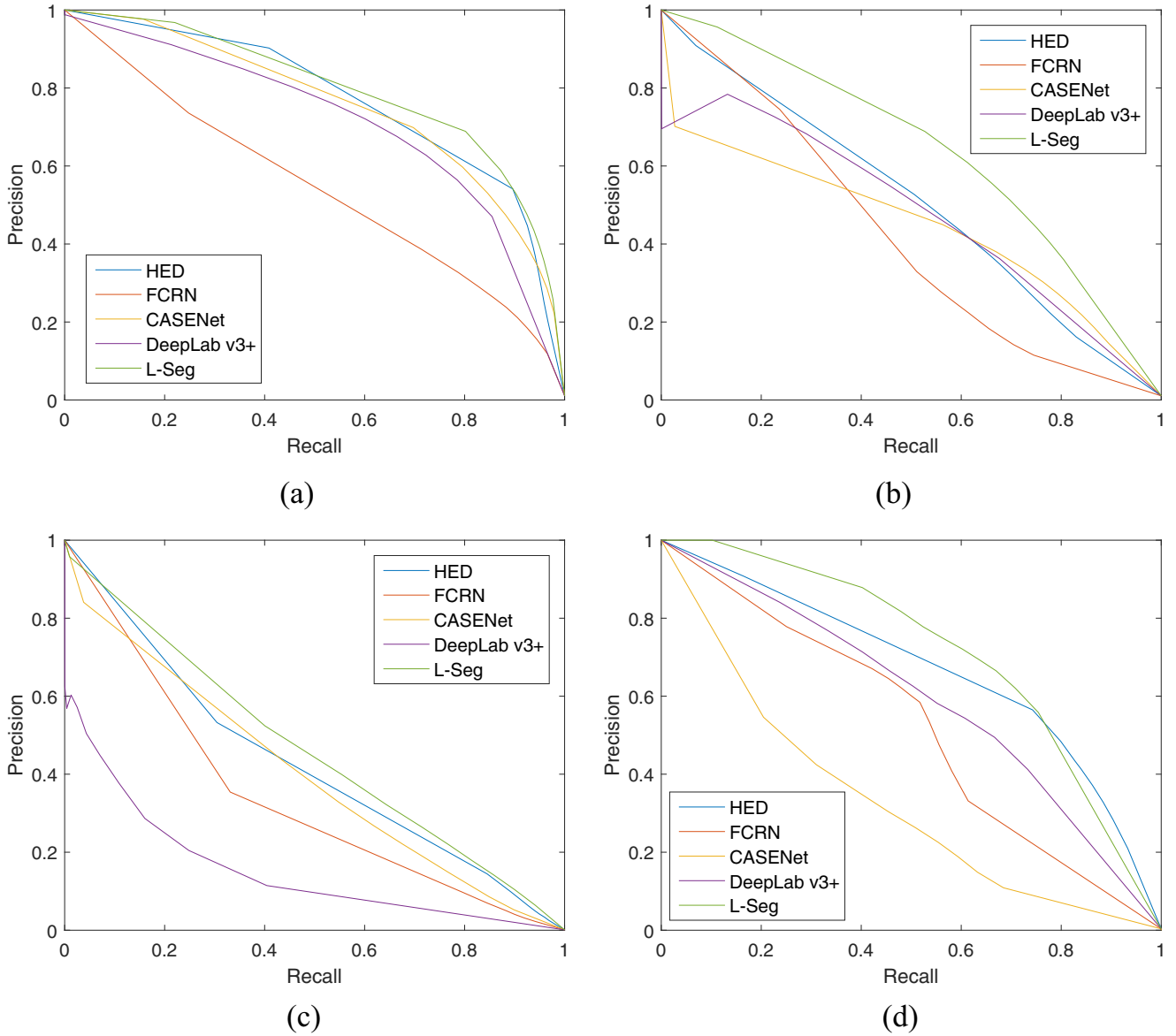


Fig. 3. (a)–(d): The PR curves for EX, HE, MA, and SE on IDRiD dataset.

Table 9
Performance comparison of exudate segmentation on e-optha EX dataset.

Method	Sensitivity	PPV	F-score
Imani et al. [11]	0.8032	0.7728	0.7877
Liu et al. [27]	0.76	0.75	0.76
L-Seg	0.8417	0.8345	0.8381

5. Discussion

5.1. Parameter setting of λ

According to the definition of MCB loss, the hyper-parameter λ is used to determine the number of background pixels. Given a specific application, it is hard to determine the optimal value of λ . However, there are some experience about the setting of λ . First, when λ is set large enough, all background pixels are selected, which means MCB loss degenerates to MC loss. This phenomenon should be avoided. Second, the value of λ highly depends to the percentage of positive pixels. Therefore, we set λ equals 10 in our experiments. One thing worth noting is that the MCB loss

is designed for class-imbalance task, so it may not behave well in class-balance task.

5.2. Limitations of L-Seg

L-Seg is proposed as an end-to-end framework for multi-lesion segmentation of fundus image. In our experiment, L-Seg achieves much higher mAUC than top 3 teams of IDRiD challenge, but we did not achieve the highest AUC value on single-lesion segmentation. In addition, our proposed MCB loss could only alleviate the problem of mis-classification to a certain extent, there are still many misclassified pixels.

5.3. Future work

Alleviating misclassification in lesion segmentation is the most critical and core problem. Existing publicly available dataset contains a small number of annotated images, which may be enough for traditional method but not enough for deep neural network. Unfortunately, IDRiD is the only publicly available dataset that provides four types of lesions annotation. Following this direction, our

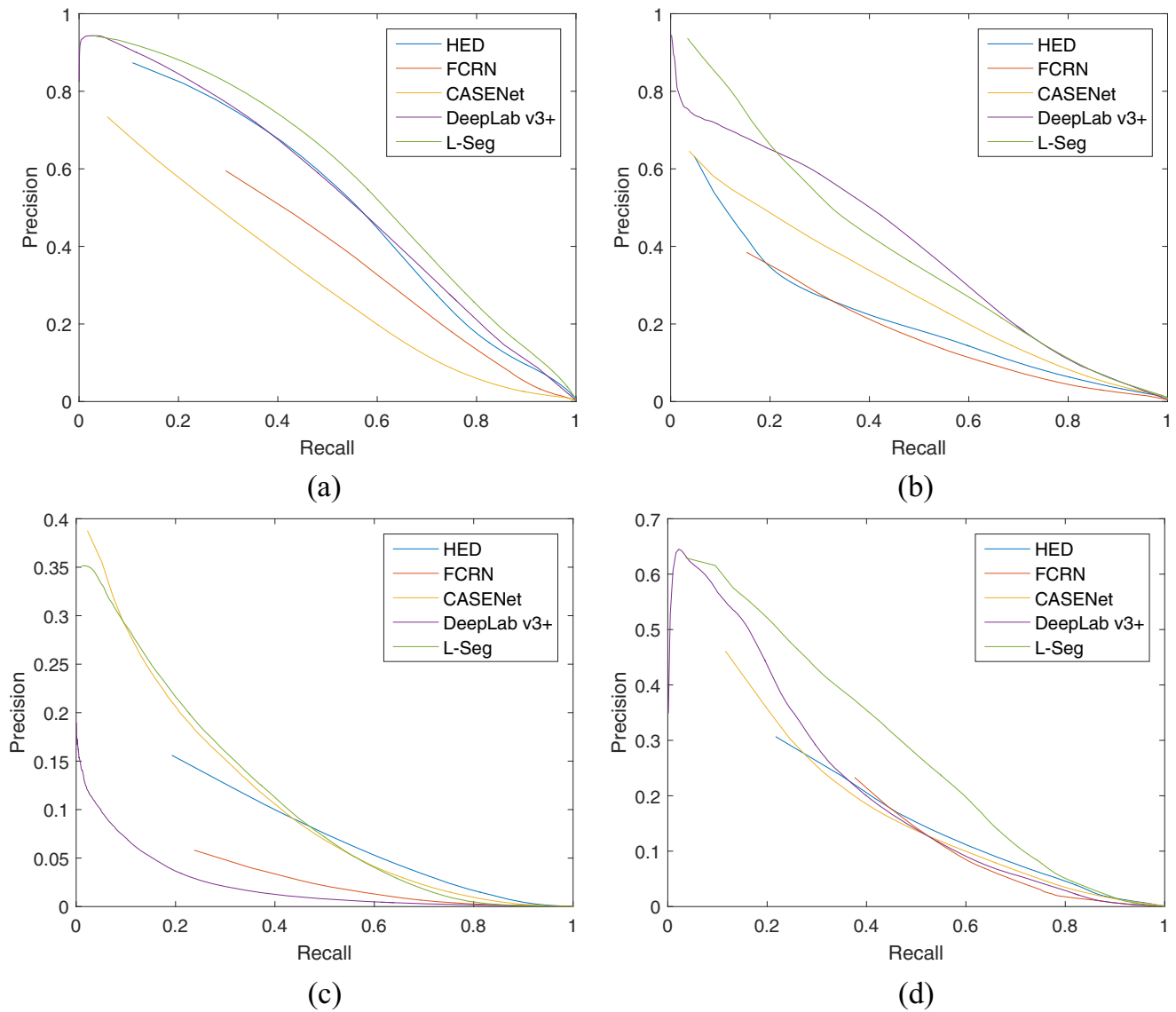


Fig. 4. (a)–(d): The PR curves for EX, HE, MA, and SE on DDR dataset.

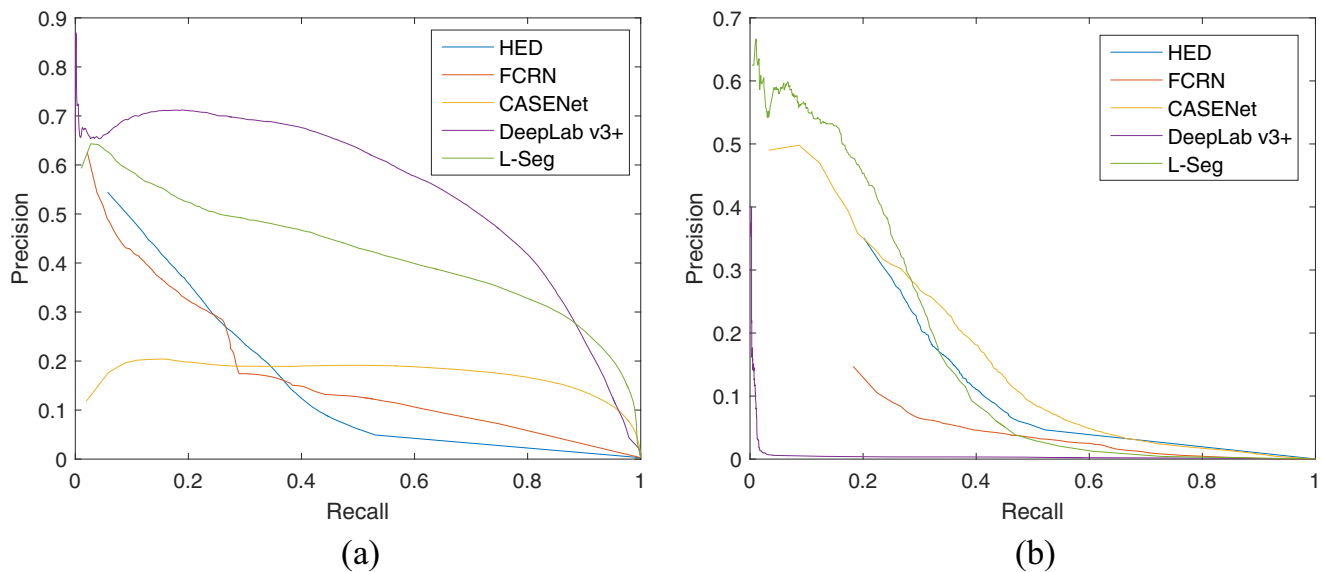


Fig. 5. (a),(b): The PR curves for EX, MA on e-optha dataset.

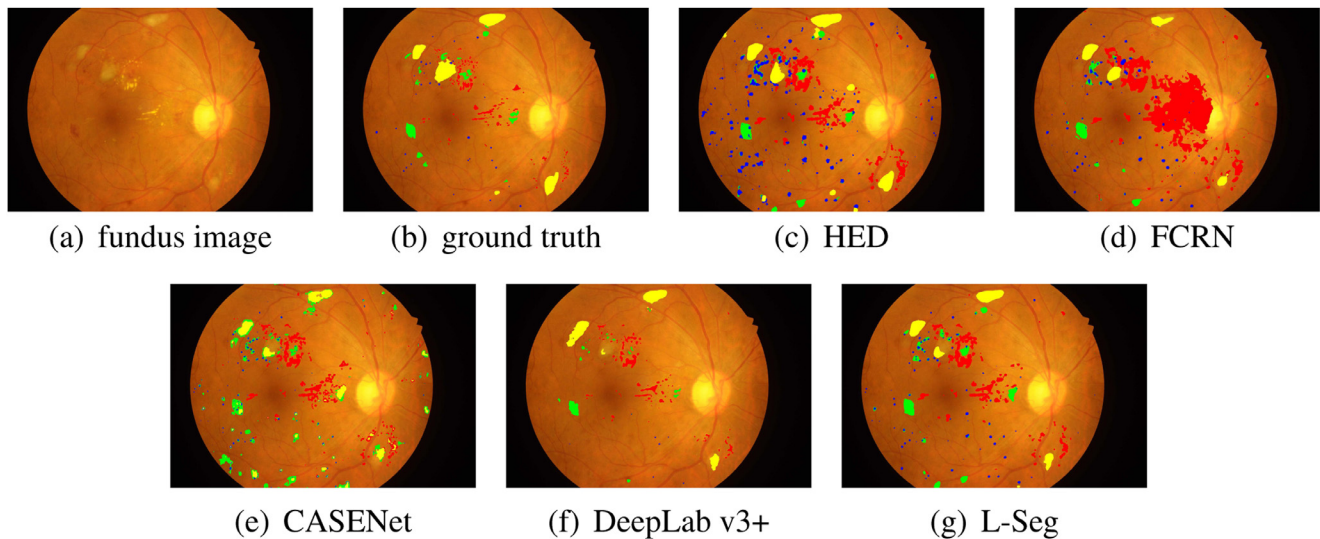


Fig. 6. The binary segmentation results of four kinds of lesions on IDRiD dataset. Red markings denote EX, green markings denote HE, blue markings denote MA and yellow markings denote SE.

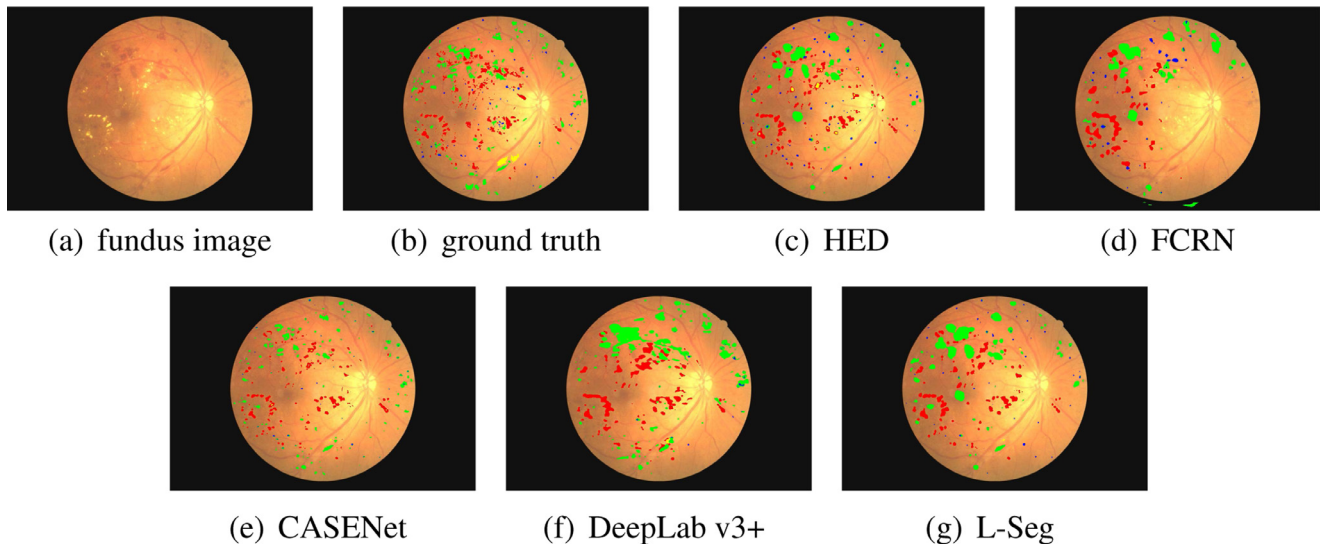


Fig. 7. The binary segmentation results of four kinds of lesions on DDR dataset. Red markings denote EX, green markings denote HE, blue markings denote MA and yellow markings denote SE.

future work includes designing anti-noise deep learning model, publish a fundus image dataset that contains thousands of annotated images and adopting domain adaption method.

6. Conclusion

The segmentation of EX, SE, MA and HE is challenging. In this paper, we proposed a novel framework named L-Seg for multi-lesion segmentation of fundus image. To the best of our knowledge, it is the first semantic segmentation network that can discriminate hard exudates, soft exudates, haemorrhages, and microaneurysms simultaneously and automatically. In addition, we proposed weighted fusion method and multi-channel bin loss for improving segmentation accuracy and alleviating mis-classification problem. We have evaluated L-Seg on IDRiD, DDR and e-optha datasets. Experimental results show that L-Seg behaved better than two multi-classification networks (i.e., DeepLab v3+, CASENet) and two single-lesion segmentation networks (i.e., HED, FCRN). Also, L-Seg shows superior performance compared with traditional methods. Moreover, L-Seg shows competitive performance compared with top 3 teams of IDRiD challenge. However, there are

still serious misclassification problems in small lesion segmentation, and more researches are necessary to further improve the performance.

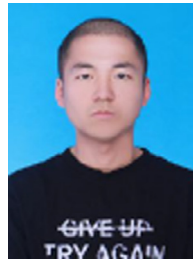
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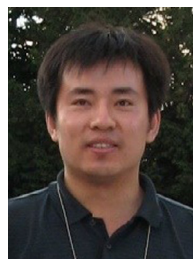
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